VitaTwin AI — System Architecture

Predict. Simulate. Prevent. | *Happiest Health Digital Twin Challenge 2026*

Static / Historical Data

- Synthetic T2D-relevant EHR
- Demographics, HbA1c, diagnosis,
- duration, medications

Dynamic / Real-Time Data

- Wearable + CGM stream (5-min)
- Glucose, HR, HRV, steps, sleep
- Rule-based physiological simulator

Data Provenance Tag

- Every row: data_source column
- synthetic vs. real_proxy (future)
- See docs/dataset.md

Data Pipeline

- Validation → cleaning → timestamp alignment → patient mapping → missing-value handling → feature engineering

Digital Twin Engine (per-patient, stateful — VitaTwinState)

Fused Twin State

- demographics · historical
- dynamic · derived features
- personal baseline deviation

Prediction Engine

- Regression: glucose @ +2h
- Classification: excursion risk
- RandomForest (beat linear baseline)
- Test MAE 11.0 | ROC-AUC 0.92

Risk + Explainability

- Risk: LOW / MODERATE / HIGH
- Trend + confidence
- SHAP contributing factors

What-If Simulator

- Forked state (no mutation)
- Baseline vs. simulated
- Labelled hypothetical only

SQLite Database

- patients · ehr_records · sensor_readings
- twin_states · predictions · simulation_runs

FastAPI Backend

- /patients/{id}/twin · /prediction · /risk
- /sensor-update · /simulate

Streamlit Doctor Dashboard

- Overview · Twin · Prediction · Why? · What-If · Live Twin

Research/educational prototype — outputs are experimental model estimates, not medical diagnoses.